From Pixels to Patches: Pooling Strategies for Earth Embeddings

Isaac Corley, Caleb Robinson, Inbal Becker-Reshef, and Juan M. Lavista Ferres

***Abstract*—Geospatial foundation models (GFMs) increasingly expose pixel-level embedding products that can be reused across tasks without access to the underlying encoder. Downstream tasks with patch- or region-level labels require aggregating these dense pixel embeddings into a single representation—yet the choice of aggregation operator is rarely examined. We show that a single structural quantity, the ratio N/D of pixels per region to embedding dimension, predicts which pooling strategy will succeed: when $N \gg D$, distributional statistics capture intra-patch heterogeneity that mean pooling discards; when $N \approx D$, second-order statistics become ill-conditioned and first-order methods dominate. To support this analysis we introduce *EuroSAT-Embed*, pre-computed pixel-level embeddings from three GFMs (AlphaEarth, OlmoEarth, Tessera) for 27,000 EuroSAT patches, and benchmark 13 pooling operators under random and geographically disjoint splits. On EuroSAT (N=4096 $\gg D$), distributional statistics (*stats*, *signed NC-GeM*) reduce the spatial generalization gap by up to 65% relative to mean pooling. On PASTIS ($\sim$85K agricultural parcels, median N=196), where $N/D \approx 1.5$–3, covariance pooling collapses from 91% to 58.7% F1 while *signed NC-GeM* remains the most robust method (AEF: 66.4% F1). The N/D ratio gives practitioners a principled decision rule: use *mean+std* or *signed NC-GeM* as defaults; add higher-order statistics only when $N \gg D$.**

***Index Terms*—Geospatial foundation models, embeddings, pooling, land cover classification, crop classification, spatial generalization.**

I. Introduction

A typical agricultural parcel in France covers 1.96 ha—196 pixels at 10 m resolution. A standard EuroSAT land-cover patch, by contrast, contains 4,096 pixels. Both settings require aggregating pixel embeddings from a geospatial foundation model (GFM) into a single patch representation, yet the effective sample size N differs by a factor of 20. We show this difference determines which pooling method wins.

GFMs now expose dense pixel-level embeddings as downloadable data products [1]–[5], enabling practitioners to reuse them without access to the underlying encoder. Many downstream tasks—crop mapping [6]–[8], building detection, land-cover classification—assign a single label to a region rather than a pixel, requiring a post-hoc aggregation step (Fig. 1). This *input-label resolution mismatch* [9] is ubiquitous: EuroSAT [10] uses 64×64 patches, PASTIS [11] uses irregular agricultural parcels. The default aggregation—mean pooling—collapses the embedding distribution to a single first-moment summary, discarding within-region variability that may carry class signal.

Pooling strategies are well-studied in image retrieval—generalized mean (GeM) [12], VLAD [13], NetVLAD [14], and Fisher Vectors [15]—and in temporal audio modeling [16], but remain largely unexplored for GFM embeddings over irregular-polygon regions. The key challenge specific to this setting is that N varies by three orders of magnitude across region types and sizes. For small parcels where $N \approx D$, the sample covariance matrix is rank-deficient, making second-order statistics unreliable. For large fixed-size patches where $N \gg D$, second-order statistics are well-estimated and outperform mean pooling by a large margin. No prior work characterizes this transition or gives practitioners a decision rule.

We study the post-hoc pooling step across the full N/D spectrum. Using pixel embeddings from three GFMs on two benchmarks, we benchmark 13 training-free and 2 fitted pooling methods and measure how method rankings shift with effective sample size. Our contributions are:

- We show that N/D is a sufficient decision criterion for pooling method selection, explaining rank reversals across the full method leaderboard between EuroSAT (N/D=64) and small PASTIS parcels ($N/D\approx$1.5) (Sec. V).
- We benchmark 13 training-free and 2 fitted pooling operators on EuroSAT (3 GFMs, random and spatial splits, kNN and linear probes), finding that distributional statistics reduce the geographic generalization gap by up to 65% vs. mean pooling (Sec. V).
- We release *EuroSAT-Embed*—pre-computed pixel-level embeddings from three GFMs—and all benchmark code, providing the first controlled study of post-hoc pooling for GFM embedding products (Sec. II).

II. Data and Embeddings

EuroSAT. We use the 10-class EuroSAT land-cover dataset [10], which contains 27,000 Sentinel-2 patches at 64×64 pixels. We evaluate on both the standard random split [17] and a spatial split [18], [19] that partitions samples by longitude to reduce train-test leakage via spatial autocorrelation.

PASTIS. We use the PASTIS panoptic segmentation dataset [11], which contains 2,433 Sentinel-2 time-series patches covering France with instance-level annotations of $\sim$85,000 agricultural parcels across 18 crop classes. Parcels are irregular polygons extracted from instance masks; parcel

Isaac Corley is with Taylor Geospatial, USA (e-mail: isaac.corley@taylorgeospatial.org).

Caleb Robinson, Inbal Becker-Reshef, and Juan M. Lavista Ferres are with Microsoft AI for Good Research Lab, USA.

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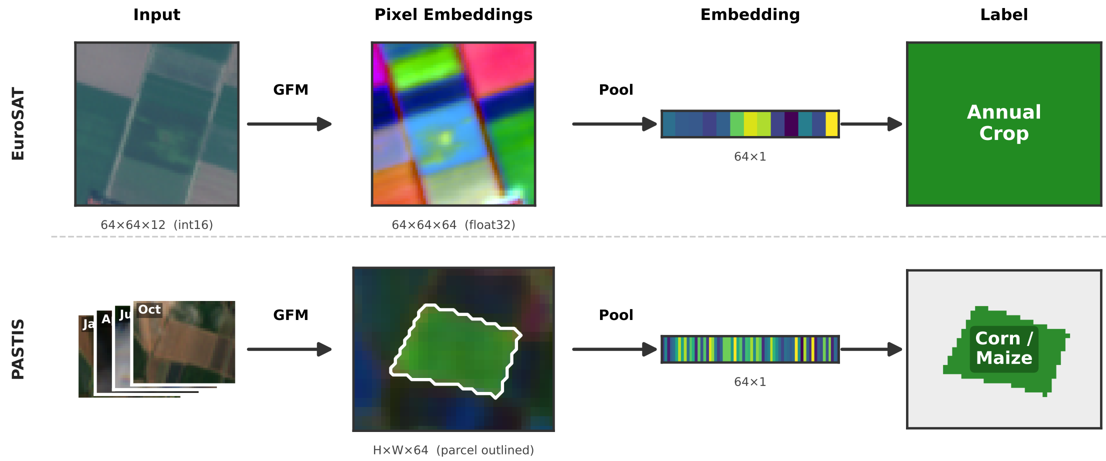

Fig. 1. **Embedding pipeline for EuroSAT (top) and PASTIS (bottom).** In both settings, a GFM encodes every pixel independently and a pooling operator aggregates those pixel vectors into one patch or parcel representation. *Top:* A 64×64 Sentinel-2 patch is encoded pixel-wise; all 4,096 pixel embeddings are pooled into a single patch vector for land-cover classification. *Bottom:* Multi-temporal Sentinel-2 frames are encoded per-pixel; only pixels inside the parcel boundary (contour shown on the PCA pseudo-RGB embedding) are pooled into a single parcel vector for crop-type classification. The number of pixels per parcel N varies from 5 to 3,440, making the ratio N/D a key design parameter.

size varies from 5 to 3,440 pixels (median 196). We use the standard 5-fold CV protocol (folds 1–4 train, fold 5 test). Parcels with fewer than 5 pixels are excluded.

EuroSAT-Embed. We construct pixel-level embedding datasets from AlphaEarth [20] (AEF, 64-d int8), OlmoEarth-Nano [21] (128-d float32, T=12 subsampled time steps), and Tessera [22] (128-d float32, annual global). AEF and OlmoEarth embed patch context and emit pixel-wise vectors; Tessera encodes per-pixel time-series without spatial context. The same three embedding sources are applied to PASTIS parcels. We study pooling behavior, not encoder comparison.

III. Pooling Methods

We benchmark training-free operators that compress an $H\times W\times D$ embedding tensor (N=HW pixels) into a d-dimensional vector.

First-order statistics (D-d): *mean* ($\mu = \frac{1}{N}\sum_i x_i$), *max*, generalized mean (*GeM*) [12], and *center-weighted mean* (down-weighting boundary pixels).

Distributional statistics ($2D$–$5D$-d): *std*, *mean+std* ($2D$-d), *mean+max* ($2D$-d), *stats* (min/max/mean/std, $4D$-d), *percentiles* ($5D$-d), *median+IQR* ($2D$-d), and *covariance* [23] (upper triangle, $D(D+1)/2$-d).

Signed non-cancelling GeM (NC-GeM) ($2D$-d): applies the power-mean separately to positive and negative activations before concatenating, preventing sign cancellation. This is the key method for variable-size regions: unlike covariance, it does not require $N > D$ to produce well-conditioned estimates.

Parametric methods: *PCA* (mean-pooled then projected) and *Bag of Visual Words* (*BoVW*) [24] (mini-batch k-means histogram). Both operate post-hoc but fit an additional stage on the training split.

IV. Experimental Setup

We evaluate two probes: kNN (k=5, cosine distance) and multinomial logistic regression with regularization C selected by 3-fold CV on the training split. We standardize before linear probing [25]. EuroSAT results appear in Tables I and II under random and spatial splits; PASTIS results (F1-macro) appear in the right columns (OE = OlmoEarth-Nano, Tess = Tessera).

V. Results

The N/D ratio determines the winning strategy. The central finding, shown across both benchmarks, is that the ratio of pixels per region N to embedding dimension D governs which pooling method wins. EuroSAT patches have N=4,096 and $D\in\{64, 128\}$, giving $N/D \geq 32$: second-order statistics are well-estimated. PASTIS parcels have a median N=196; for OlmoEarth and Tessera (D=128), $N/D \approx 1.5$—near rank-deficiency. This single structural quantity explains every major finding below.

When $N \gg D$: distributional statistics win. On EuroSAT spatial splits, *stats* pooling reaches 91–95% accuracy (avg. 93.3%) vs. 81–92% for *mean* (avg. 86.3%), a +7.0 pp gain (Table I). *Covariance* leads on random splits for two of three encoders (OlmoEarth: 97.2%, Tessera: 97.6%). Fig. 3 shows that distributional methods are simultaneously more accurate and more robust: *mean* pooling drops 9.6 pp from random to spatial splits; *stats* drops only 3.4 pp—a 65% reduction in the generalization gap. *Mean+max* (4.0 pp) and *covariance* (4.8 pp) also exhibit small gaps. kNN results (Table II) show consistent trends, with *stats* leading on spatial splits (avg. 88.0%) and *center-weighted mean* leading on random splits (avg. 95.3%).

When $N \approx D$: second-order statistics collapse. On PASTIS, *covariance* drops from 91.0% on EuroSAT to 58.7% F1 on AEF—a 32 pp collapse—because many parcels have $N < D$, making the sample covariance matrix rank-deficient. *Stats* similarly underperforms. In contrast, *signed NC-GeM* (AEF: 66.4%, OlmoEarth: 52.7%) and *mean+std* (OlmoEarth: 54.9%) remain robust. First-order methods (*mean*, *center-weighted*) are now competitive, unlike on EuroSAT. Tessera's

TABLE I
LINEAR PROBE RESULTS ON EUROSAT (ACCURACY, %) AND PASTIS (F1-MACRO, %). BEST PER COLUMN IN **BOLD**, SECOND-BEST IN *italics*. GAP = ACCURACY DROP FROM RANDOM TO SPATIAL SPLIT. DIM = OUTPUT EMBEDDING SIZE RELATIVE TO ENCODER DIMENSION D. SIGNED NC-GEM AND STATS ARE HIGHLIGHTED AS RECOMMENDED METHODS. *OLMOEARTH-NANO; OE = OLMOEARTH; TESS = TESSERA.

		Spatial split				Random split					PASTIS			
Pooling	Dim	AEF	OE*	Tess	Avg	AEF	OE*	Tess	Avg	Gap↓	AEF	OE*	Tess	Avg
Mean	D	81.1	91.8	86.0	86.3	95.3	96.4	96.1	95.9	9.6	64.5	54.7	*49.0*	56.1
Std	D	77.1	90.5	85.4	84.3	90.9	94.8	95.3	93.7	9.3	49.0	22.8	40.3	37.4
GeM	D	83.2	91.7	87.3	87.4	95.5	95.9	96.4	95.9	8.5	62.8	49.7	48.1	53.5
Center-Weighted	D	85.7	91.9	87.6	88.4	96.1	96.8	96.1	96.3	7.9	65.3	*54.8*	48.9	*56.3*
Max	D	86.1	90.6	89.8	88.8	92.9	92.9	93.7	93.2	4.3	57.4	38.8	45.5	47.2
Signed NC-GeM	$2D$	83.1	93.6	87.9	88.2	95.5	96.4	96.4	96.1	7.9	**66.4**	52.1	49.0	55.8
Mean+Std	$2D$	84.2	93.4	*94.8*	90.8	*96.1*	96.6	97.1	96.6	5.8	*65.5*	**54.9**	48.7	**56.4**
Mean+Max	$2D$	*89.1*	93.3	94.4	*92.3*	96.1	96.3	96.5	96.3	*4.0*	65.0	54.4	48.8	56.1
Median+IQR	$2D$	81.7	93.2	88.4	87.7	94.8	95.9	95.9	95.5	7.8	64.3	50.8	48.4	54.5
Percentiles	$5D$	85.6	*93.8*	91.2	90.2	95.9	*97.0*	97.0	96.6	6.5	65.3	52.4	**49.4**	55.7
Stats	$4D$	**91.0**	**94.0**	**94.9**	**93.3**	**96.2**	96.6	*97.3*	*96.7*	**3.4**	65.0	54.0	48.4	55.8
Covariance	$D(D+1)/2$	89.1	93.4	93.7	92.1	95.8	**97.2**	**97.6**	**96.8**	4.8	58.7	42.4	47.0	49.4
Parametric (fit on train split)														
PCA	64	79.7	91.6	80.8	84.0	95.2	95.6	95.0	95.3	11.3	64.4	47.9	48.2	53.5
BoVW	128	88.9	88.2	88.7	88.6	94.4	94.5	96.3	95.1	6.5	44.7	23.0	15.5	27.8

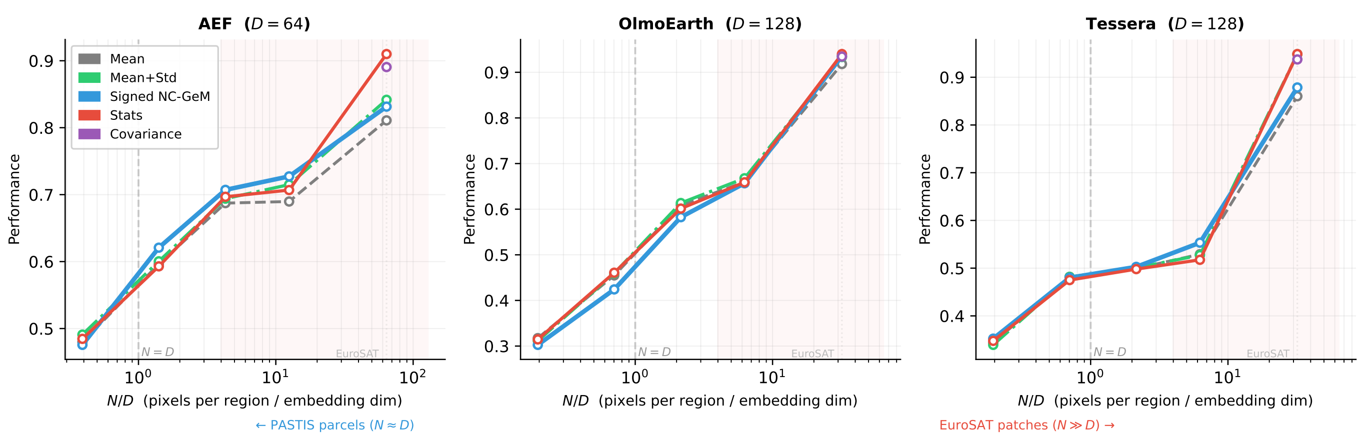

Fig. 2. **Pooling performance as a function of N/D, the ratio of pixels per region to embedding dimension.** Each panel shows per-bin F1 (PASTIS, four parcel-size bins) and accuracy (EuroSAT) plotted against N/D on a log scale for one GFM. The dashed vertical line marks $N{=}D$. To the left, where parcels are small and $N \approx D$, covariance (purple) collapses because its sample estimate becomes rank-deficient, while mean+std and signed NC-GeM remain strong. Moving right toward EuroSAT patches where $N \gg D$, covariance recovers and leads. Signed NC-GeM (blue) stays competitive across the full range—the only method to do so.

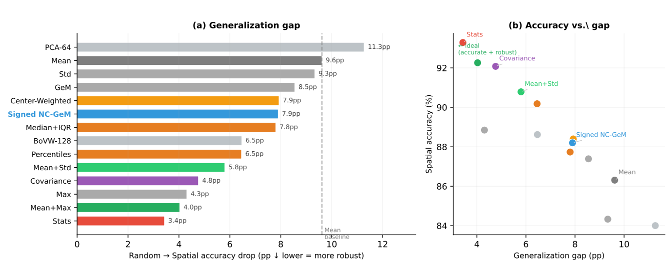

Fig. 3. **Distributional pooling is more accurate and more robust under spatial shift.** (a) Accuracy drop when moving from a random to a geographically disjoint split, averaged across three GFMs—shorter bars are better. Mean pooling drops 9.6 pp; stats drops only 3.4 pp, a 65% reduction. (b) Spatial accuracy vs. generalization gap per method. Methods in the upper-left are both accurate and robust; distributional methods cluster there while mean pooling sits lower-right.

comparatively low scores across all methods (best: 49.4%) reflect its annual composites, which lack the seasonal phenological signal needed to discriminate crop types.

Region-size breakdown confirms the N/D prediction. Fig. 4 ranks methods within four parcel-size bins averaged across all three embedding sources. For tiny parcels ($N < 50$), *mean+std* and *center-weighted mean* tie at avg. rank 2.0; for medium (150–500 px) and large ($>$ 500 px) parcels, *signed NC-GeM* ranks first (2.7 and 2.3). *Covariance* ranks last in the tiny and small bins and first in the large bin—a complete rank reversal that the N/D ratio predicts exactly. *Signed NC-GeM* is the only method that stays in the top-3 across all bins, making it the practical default for variable-size regions.

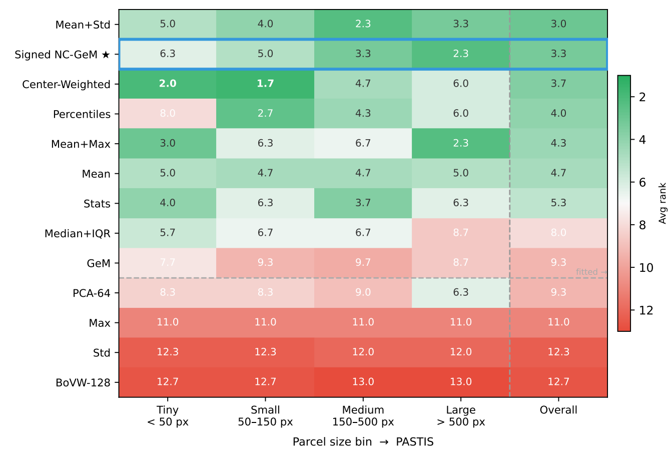

Fig. 4. **Method rankings flip as parcel size grows.** Each cell shows the average rank across AEF, OlmoEarth, and Tessera within each parcel-size bin—green is rank 1 (best), red is last. Covariance (bottom-right) ranks last for tiny parcels and first for large ones, a complete reversal explained by the N/D ratio. Signed NC-GeM (★, highlighted in blue) is the only method that stays in the top 3 across all bins. The dashed vertical line separates per-bin results from the overall ranking.

A. PASTIS Crop Classification

Fig. 4 summarizes the per-bin ranking. AEF achieves the highest overall F1 because its $D{=}64$ gives a more favorable N/D ratio at small parcel sizes than OlmoEarth or Tessera ($D{=}128$). This is consistent with the N/D prediction: AEF's smaller embedding space makes second-order statistics viable at smaller parcel sizes.

VI. Discussion

Recommendation. The N/D ratio gives practitioners a decision rule that does not require dataset-specific tuning. When $N \gg D$ (large fixed-size patches, *e.g.* 64×64 at 10 m), use *stats* or *covariance* for peak accuracy; *stats* is more robust to distribution shift. When $N \approx D$ or regions are variable-size (parcels, fields, buildings), use *signed NC-GeM*: it is the only method in the top-3 across all parcel sizes. *Mean* remains a useful lower bound. Notably, accuracy and robustness are aligned: *stats* and *signed NC-GeM* are simultaneously more accurate and more robust than *mean*, with no accuracy–robustness tradeoff.

Why distributional statistics help. Mean pooling collapses each region to a first-moment summary, discarding intra-region heterogeneity. A residential neighborhood and an industrial zone may share a mean embedding while differing in variance: the mean erases that distinction; *stats* or *signed NC-GeM* preserves it. This signal is reliable only when N is large enough to estimate higher moments—which is precisely what N/D measures.

Practical applications. The N/D framework maps directly onto GIS zonal statistics workflows [26], [27], where raster pixel values are aggregated over vector polygons (agricultural fields, building footprints, administrative units, ecological zones). As GFM embedding products become available at continental scale [1], practitioners who run polygon-level inference—crop monitoring [6], habitat classification, urban mapping—can apply our decision rule at inference time with no additional training: use *signed NC-GeM* for variable-size or small polygons, *stats* or *covariance* for large fixed chips.

Limitations. We evaluate on EuroSAT (10 classes) and PASTIS (18 classes) with three embedding sources; broader coverage (*e.g.*, BigEarthNet [28], fMoW [29], multi-label settings) would strengthen conclusions. OlmoEarth is evaluated with $T{=}12$ subsampled time steps. We focus on training-free pooling; learned aggregation [30] may improve further but requires encoder access.

VII. Conclusion

We showed that the pixel-to-dimension ratio N/D is a principled decision criterion for post-hoc pooling of geospatial

TABLE II
kNN (k=5) results on EuroSAT (accuracy, %) and PASTIS (F1-macro, %). Best per column in **bold**, second-best in *italics*. Gap = accuracy drop from random to spatial split. Dim = output embedding size relative to encoder dimension D. Signed NC-GeM and Stats are highlighted as recommended methods. *OlmoEarth-Nano; OE = OlmoEarth; Tess = Tessera.

		Spatial split				Random split					PASTIS			
Pooling	Dim	AEF	OE*	Tess	Avg	AEF	OE*	Tess	Avg	Gap↓	AEF	OE*	Tess	Avg
Mean	D	85.5	87.8	84.7	86.0	94.6	94.8	94.9	94.8	8.8	56.5	**41.1**	**43.8**	**47.1**
Std	D	78.1	81.7	84.0	81.3	93.7	93.3	94.1	93.7	12.4	48.2	14.4	37.2	33.3
GeM	D	85.7	86.6	85.4	85.9	94.6	93.3	95.1	94.3	8.5	54.2	36.6	43.3	44.7
Center-Weighted	D	86.0	87.8	85.5	86.4	**95.4**	**95.0**	95.5	**95.3**	8.9	*57.1*	*40.3*	*43.7*	*47.0*
Max	D	80.5	85.5	80.8	82.3	90.5	91.4	92.5	91.5	9.2	46.0	27.6	39.3	37.6
Signed NC-GeM	$2D$	85.7	87.9	85.4	86.3	94.7	*95.0*	95.1	94.9	8.6	**57.7**	38.7	42.5	46.3
Mean+Std	$2D$	86.4	*88.3*	86.7	*87.1*	94.9	94.9	95.3	*95.0*	7.9	56.7	39.7	43.5	46.6
Mean+Max	$2D$	85.3	87.9	86.9	86.7	94.7	94.4	**95.8**	94.9	8.2	53.0	35.2	42.7	43.6
Median+IQR	$2D$	86.0	86.0	85.7	85.9	94.1	94.0	94.4	94.1	8.2	55.7	35.7	43.3	44.9
Percentiles	$5D$	*86.7*	87.5	86.2	86.8	94.6	94.9	95.4	95.0	8.2	56.6	39.1	43.0	46.3
Stats	$4D$	**86.9**	87.6	*89.3*	**87.9**	*95.0*	94.4	*95.6*	95.0	7.1	52.1	34.3	42.5	43.0
Covariance	$D(D{+}1)/2$	84.4	84.8	**89.8**	86.4	93.4	94.3	94.5	94.1	7.7	47.6	22.4	39.6	36.6
Parametric (fit on train split)														
PCA	64	85.3	**88.4**	86.1	86.6	93.5	94.1	92.7	93.5	*6.8*	56.2	38.8	43.3	46.1
BoVW	128	84.0	82.1	87.4	84.5	90.7	90.3	90.2	90.4	**5.9**	43.6	22.1	14.1	26.6

foundation model embeddings, unifying results across two benchmarks and three GFMs. When $N \gg D$, distributional statistics reduce the spatial generalization gap by up to 65% and improve classification accuracy by up to 7 pp over mean pooling; when $N \approx D$, second-order statistics collapse and *signed NC-GeM* is the most robust alternative. We recommend *signed NC-GeM* as the default for variable-size regions and *stats* for large fixed-size patches.

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